ORLANDO HEALTH
1414 Kuhl Ave.
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DEPARTMENT GUIDELINE

Guideline Title: **Chest Tube Management**	Guideline Number: **GUID-3203-PR005**

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature:	Signature:
Department Numbers: 3203	

I. **PURPOSE:** To maintain the chest tube and device used to evacuate air and fluid from the pleural cavity and/or to restore normal dynamics of pulmonary function.

II. **DEFINITIONS:**

III. **DEPARTMENT GUIDELINE:**

a. INDICATIONS: A chest tube in place as a result of the following- hemothorax, pneumothorax, chylothorax (lymphatic fluid leak), or pleural effusion

b. CONTRAINDICATIONS: None

c. PROCEDURE: Assess the patient, tube, and chamber to assure proper functioning.

  i. Patient: Assess for respiratory distress, lung sounds, subcutaneous emphysema, vital signs, chest, x-ray

  ii. Tube:

    1. Location of the tube: Right/Left Pleural or Mediastinal
    2. Drainage holes should not be visible external to the chest wall. If they are visible:
    a. Cover with occlusive dressing or consider replacement of tube.
    b. Never advance a chest tube once the sterile field has been removed.
    3. Make sure connections are tight.
    4. Tape the tube to the patient very well
    5. Look for clots in tube. If clots are present, milk but do not strip tube.

  iii. Unit: Water filled unit (i.e. Atrium):

    1. It should be about 12 inches below the patient and below the level of patient’s chest when transferring patient to aircraft.
    2. Confirm ordered amount of pressure. Ensure the suction source vacuum is at -80 mmHg or higher. Continue suction en route to receiving facility.
    3. Assess water seal. Water seal chamber should be at 2 cm.
    4. Bubbling:
      a. Gentle bubbling is the goal.

b. Absent is appropriate if the bubbling unit is not hooked to suction and the remainder of patient assessment is stable.
c. Vigorous is indicative of excessive air leak or suction is set too high.
d. Recheck all connections

5. Tidaling:
   a. With respirations is good
   b. Decreased could mean an obstruction
   c. Excessive could mean distress.
6. Fluid level should be per manufacturer's specs.

iv. Waterless unit:
   1. Dial in the pressure to 20 cm of water
   2. Attach to suction as necessary.

v. Output:
   1. Note output since insertion (past 8 hours if in place > 24 hours) + Output in transport.
   2. Consider blood product replacement.

vi. Use a Y-connector to connect to chambers to one suction unit if necessary.

vii. Avoid dependent loops in the tubing.

viii. Do not clamp the drainage tubing at any time unless you are about to change the drainage system or assessing for the location of an air leak.

ix. If the drainage system is knocked over, gently tip the drain to the right to return the fluid to the correct column.

x. Heimlich valves are for pneumothorax only. If Heimlich valve present, ensure valve is connected to tube securely

xi. Frequently reassess patient to ensure airway/ventilation remains intact. Monitor for signs of tension or respiratory distress.

IV. **DOCUMENTATION**: EMS charts

V. **REFERENCES:**